Project Proposal: Quantitative Risk Modeling & Game-Theoretic Analysis of Nepal's "10-Kitta" IPO Lottery System

Target Track: Quantitative Research / Trading

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Abstract

This project proposal details a quantitative research framework designed to model, simulate, and evaluate the expected return, downside risk, and game-theoretic equilibrium of multi-account retail bidding strategies within the Nepal Stock Exchange (NEPSE) Initial Public Offering (IPO) lottery system. By integrating probability theory, non-cooperative game theory, Gaussian copulas, and stochastic Monte Carlo simulations, this study isolates the precise market parameters under which retail liquidity allocation shifts from alpha generation to net-negative yield due to hidden execution frictions and adverse selection.

1 Executive Summary & Objective

Under the regulatory framework enforced by the Central Depository Services and Clearing Ltd. (CDSC) in Nepal, oversubscribed retail IPOs are distributed via a uniform random lottery in fixed lots of 10 shares (10 Kitta, par value NPR $1,000$). To maximize aggregate win probability, retail investors deploy an account-splitting strategy by filing parallel applications across n family Demat accounts.

The primary objective of this project is to construct an end-to-end quantitative pipeline in Python that:

1. Models the mathematical Expected Value (EV) of multi-account lottery participation under fixed operational frictions.
2. Derives the game-theoretic Nash Equilibrium where aggregate market account dilution renders the marginal return of additional accounts net-negative.
3. Mitigates the *Winner's Curse* by conditioning demand and listing returns on a composite issuer Prestige Score (Q).
4. Executes a $10,000$-run Monte Carlo simulation to evaluate portfolio risk metrics (Sharpe Ratio, Value-at-Risk, Max Drawdown).

2 Theoretical & Mathematical Framework

2.1 Single-Account Expected Value (EV_{single})

Let S denote the total public share volume issued, yielding $L = \frac{S}{10}$ total winning lots. For an aggregate market demand of A total applicants, the single-account win probability P_{win} is:

$$P_{\text{win}} = \min\left(1, \frac{L}{A}\right) \tag{1}$$

Factoring in direct application fees (C_{ASBA}), prorated annual maintenance fees (C_{annual} across M annual issues), and the opportunity cost of locking capital $K = \text{NPR } 1,000$ at risk-free rate r over t days, the net single-account expected value is:

$$EV_{\text{single}} = P_{\text{win}} \cdot (K \cdot R_{\text{listing}}) - C_{\text{ASBA}} - \left(\frac{C_{\text{annual}}}{M}\right) - \left(K \cdot r \cdot \frac{t}{365}\right) \tag{2}$$

where $R_{\text{listing}} = \frac{P_{\text{listing}} - P_{\text{issue}}}{P_{\text{issue}}}$ represents the percentage capital gain upon secondary market listing.

2.2 Multi-Account Portfolio Dynamics & Binomial Draw

For an investor deploying n distinct family accounts, allocation outcomes follow a Binomial distribution $X \sim \text{Binomial}(n, P_{\text{win}})$. Under the assumption that $n \ll A$, portfolio expected return scales linearly:

$$EV(n) = n \cdot EV_{\text{single}} \tag{3}$$

2.3 Game-Theoretic Nash Equilibrium (A^*)

Assuming an N-player non-cooperative game where every market participant adopts n accounts, aggregate demand inflates to $A(n) = N \cdot n$. As $A(n) \to \infty$, $P_{\text{win}} \to 0$. Setting $EV_{\text{single}} = 0$ yields the critical market demand threshold A^* past which retail lottery participation becomes unviable:

$$A^* = \frac{L \cdot (K \cdot R_{\text{listing}})}{C_{\text{ASBA}} + \left(\frac{C_{\text{annual}}}{M}\right) + \left(K \cdot r \cdot \frac{t}{365}\right)} \tag{4}$$

The systemic system state reaches a Nash Equilibrium when aggregate account expansion forces $A \geq A^*$, driving the marginal Expected Value of deploying the $(n+1)$-th account below zero ($EV(n+1) - EV(n) \leq 0$).

2.4 Issuer Prestige Factor (Q) & Winner's Curse

To model adverse selection, we define a composite quality score $Q_i \in [0, 1]$ based on fundamental parameters (Net Worth per share, sector, promoter lock-in). Demand A_i and listing yield $R_{i,\text{listing}}$ are modeled as non-linear joint functions of Q_i:

$$A_i = f(Q_i) + \epsilon_A, \quad R_{i,\text{listing}} = g(Q_i) + \epsilon_R, \quad \text{where } \epsilon_A, \epsilon_R \sim \mathcal{N}(0, \sigma^2) \tag{5}$$

- **High Prestige ($Q_i \to 1$):** High returns ($R_{i,\text{listing}} \gg 0$) paired with high demand ($A_i \gg L_i$), driving $P_{\text{win}} \to 0$.
- **Low Prestige ($Q_i \to 0$):** Low demand ($A_i \approx L_i$) yielding $P_{\text{win}} \to 1$, but exposing the investor to zero or negative secondary market returns.

3 Monte Carlo Simulation Architecture

The simulation engine will execute $10,000$ stochastic iterations over a simulated 1-year horizon ($M = 20$ IPOs) within Python ('NumPy', 'SciPy', 'Pandas').

3.1 Strategy Comparison Metrics

The engine will evaluate two distinct execution models:

1. **Unfiltered Naive Strategy:** Unconditionally bid on all M IPOs across n accounts.

Table 1: Simulation Parameters and Execution Frictions

Parameter	**Symbol**	**Baseline Value / Range**
Standard Issue Size	S	$1,000,000 - 5,000,000$ shares
Application Capital per Account	K	NPR $1,000$
C-ASBA Application Fee	C_{ASBA}	NPR $5.00 - 25.00$
Annual Maintenance Overhead	C_{annual}	NPR 150.00/account
Risk-Free Savings Yield	r	6.00% p.a.
Capital Lock-up Window	t	$10 - 14$ days
Simulated Family Accounts	n	$1 \leq n \leq 10$
Monte Carlo Iterations	N_{sim}	$10,000$

2. **Quality-Filtered Strategy:** Bid only on IPOs satisfying $Q_i > \tau$, where τ is a calibrated quality cut-off threshold.

Outcomes will be evaluated using standard quantitative risk metrics:

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}, \quad \text{VaR}_{0.95} = \inf\{x \in \mathbb{R} : P(R_p \leq x) \geq 0.05\} \tag{6}$$

4 Execution Roadmap & Deliverables

- **Week 1: Data Scraping & Parsing:** Build Python scrapers ('BeautifulSoup', 'requests') to extract historical issue sizes, oversubscription ratios, and listing returns for 30–50 recent NEPSE IPOs.
- **Week 2: Math Engine Implementation:** Code vectorized mathematical modules for single/multi-account EV, equilibrium bounds A^*, and copula-based joint distribution fitters.
- **Week 3: Stochastic Simulation Engine:** Construct the full Monte Carlo pipeline in Python; compute portfolio PnL curves, Sharpe ratios, and Value-at-Risk metrics.
- **Week 4: Documentation & GitHub Packaging:** Format codebase into modular scripts ('data.py', 'models.py', 'sim.py', 'plots.py'), generate visualization plots ('Matplotlib', 'Seaborn'), and publish a research-grade repository Readme containing LaTeX derivations.